When Does Graph Propagation Help Recommendation? A Technical Review of MF-BPR and LightGCN on Dense and Sparse Implicit Feedback

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Abstract

We review and empirically compare two representative recommender-system methods: **MF-BPR**, matrix factorization trained with the Bayesian Personalized Ranking loss, and **LightGCN**, a simplified graph convolutional network over the user–item interaction graph. Both are implemented from scratch in PyTorch and evaluated under an identical full-ranking protocol on two datasets chosen to differ mainly in interaction density: MovieLens-1M (dense, 4.8%) and Amazon Reviews 2023 Video Games (144× sparser, 0.034%), with popularity and ItemKNN baselines as anchors. LightGCN outperforms MF-BPR on both datasets, but the margin grows by an order of magnitude with sparsity: +6.1% NDCG@10 on MovieLens-1M versus +43.5% on Video Games. A temporal-split robustness check sharpens this result: the sparse-data gain survives the strict "predict-the-future" protocol (+38.2%) while the dense-data gain vanishes (+0.4%), falling within the margin inflated by temporal leakage. Per-user analysis shows the gain concentrates on low-activity users, and a popularity analysis reveals that on sparse data part of the gain comes from recommending popular items at the cost of catalog coverage (62%→41%). We then propose a one-line fix: generalizing the propagation normalization to $d_s^{-\alpha} d_t^{-(1-\alpha)}$; at α=0.3 it is a strict Pareto improvement — accuracy +2.8% NDCG@10 while recovering more than half of the lost coverage. We further show a methodological pitfall: under a 300-epoch budget the layer ablation appears to exhibit "over-smoothing" beyond L=2, which vanishes entirely once every configuration is given a fair 600-epoch budget — deeper propagation simply converges more slowly. All results are produced by our own code and are reproducible from fixed seeds.

1 Introduction

Recommender systems learn user preferences from historical interactions. Matrix factorization (MF) has been the workhorse of collaborative filtering since the Netflix-Prize era: it embeds each user and item in a low-dimensional space and models affinity as a dot product, using only *direct* user–item interactions. A more recent line of work observes that the interaction data forms a bipartite graph, and that propagating embeddings over this graph injects *higher-order* collaborative signal — "users who liked the items I liked" — into the representation. LightGCN is the cleanest representative of this idea [2]: it strips graph convolution down to weighted neighborhood averaging, with no feature transforms and no nonlinearities.

This project asks a focused question: **when, and for whom, does the higher-order graph signal actually help?** We keep everything else fixed — the embedding tables, the BPR training objective, the negative sampling, the evaluation protocol — so that the *only* difference between our two methods is the propagation step. Any performance gap is then attributable to the graph. We deliberately evaluate on two datasets at opposite ends of the density spectrum, because the value of borrowed neighborhood signal should depend on how much direct signal each user already has.

Our contributions are: (i) a from-scratch, ∼100-line PyTorch implementation of both methods verified against published result ranges; (ii) a controlled comparison on dense and sparse data with per-user and popularity-level analyses that explain *where* the gains come from; and (iii) a cautionary methodological finding about training budgets in layer ablations.

2 Methods

2.1 Problem setup and the BPR objective

Let $\mathcal{U}$ and $\mathcal{I}$ be the user and item sets, and $\mathcal{O} = \{(u, i)\}$ the observed implicit interactions. Each user and item has an embedding $\mathbf{e}_u, \mathbf{e}_i \in R^d$, and a pair is scored $\hat{y}_{ui} = \mathbf{e}_u^\top \mathbf{e}_i$. At test time we rank all items the user has not interacted with in training. Both models are trained with the pairwise BPR loss [1]: for each observed (u, i) and a uniformly sampled unobserved item j,

$$\mathcal{L}_{\text{BPR}} = -\sum_{(u,i,j)} \ln \sigma(\hat{y}_{ui} - \hat{y}_{uj}) + \lambda \|\Theta\|_2^2, \tag{1}$$

which pushes interacted items above random non-interacted ones in each user's ranking.

2.2 Method 1: MF-BPR

MF-BPR uses the free embedding tables directly: $\mathbf{e}_u$ and $\mathbf{e}_i$ are the model parameters, and no other computation is performed. Its inductive bias is minimal; every user is positioned only by the items they themselves interacted with.

2.3 Method 2: LightGCN

LightGCN keeps the same tables as layer-0 embeddings $\mathbf{e}^{(0)}$ and propagates them over the interaction graph. With $\mathcal{N}_u$ the items of user u and $\mathcal{N}_i$ the users of item i,

$$\mathbf{e}_u^{(k+1)} = \sum_{i \in \mathcal{N}_u} \frac{\mathbf{e}_i^{(k)}}{\sqrt{|\mathcal{N}_u|\,|\mathcal{N}_i|}}, \qquad \mathbf{e}_i^{(k+1)} = \sum_{u \in \mathcal{N}_i} \frac{\mathbf{e}_u^{(k)}}{\sqrt{|\mathcal{N}_i|\,|\mathcal{N}_u|}}, \tag{2}$$

and the final embedding is the mean over layers, $\mathbf{e}_u = \frac{1}{L+1} \sum_{k=0}^{L} \mathbf{e}_u^{(k)}$. There are no weight matrices and no activation functions: LightGCN is deliberately a *simplification* of earlier graph collaborative-filtering models such as NGCF [3], which its authors showed helps rather than hurts for recommendation. Layer k mixes in k-hop neighbors, so L=2 already reaches other users' items. Setting L=0 recovers MF-BPR exactly — the comparison is a clean ablation by construction.

2.4 Reference baselines

To anchor the comparison we add two non-learned baselines evaluated under the identical protocol. **Popularity** ranks items by their training-set interaction count and serves the same list to every user — the "zero personalization" floor. **ItemKNN** computes cosine similarity between item columns of the binary interaction matrix (truncated to each item's top k=100 neighbors) and scores $\hat{y}_{ui} = \sum_{j \in \mathcal{N}_u} \text{sim}(i, j)$ — the classic neighborhood heuristic. Together with MF and LightGCN they form a three-generation lineage: statistical heuristic → shallow representation learning → graph representation learning.

2.5 Implementation

Both models are implemented from scratch in PyTorch (∼100 lines total). Propagation is realized as gather + `index_add` over the edge list, which runs identically on CUDA, Apple-Silicon MPS, and CPU (sparse matmul is not supported on MPS). Because LightGCN propagates the full graph at every optimization step, we use a large batch of 65,536 with lr 3×10^{-3}, giving 9× fewer propagations per epoch with convergence indistinguishable from the small-batch setting; MF uses batch 8,192 and lr 10^{-3}. Both use Adam, L_2 regularization 10^{-4}, and embedding dimension 64 unless stated. Negative sampling uses a seeded NumPy generator on the CPU, which makes runs bit-reproducible: the MF run produced *identical* test metrics on an Apple M2 and an NVIDIA L4.

3 Experimental Setup

3.1 Datasets

	MovieLens-1M	Video Games
Interactions (5-core)	999,611	814,586
Users	6,040	94,762
Items	3,416	25,612
Density (%)	4.8448	0.0336
Avg. interactions / user	165.5	8.6
Avg. interactions / item	292.6	31.8

Table 1: Datasets after preprocessing. The two datasets have interactions of the same order of magnitude but differ in density by a factor of ∼144.

MovieLens-1M [4] is the classic dense movie-rating benchmark (GroupLens Research): 1,000,209 ratings from 6,040 users. **Amazon Reviews 2023 – Video Games** [5] (McAuley Lab, UCSD) contains 4.62M reviews; gaming involves frequent repeat purchasing, so the category retains a healthy interaction graph after filtering. Both are converted to implicit feedback (every observed pair is a positive), deduplicated, 5-core filtered, and split per user 80/10/10 into train/validation/test with seed 42 (Table 1). Figure 1 shows both are long-tailed, but at very different scales: the average MovieLens user has 165 interactions, the average Video-Games user 8.6.

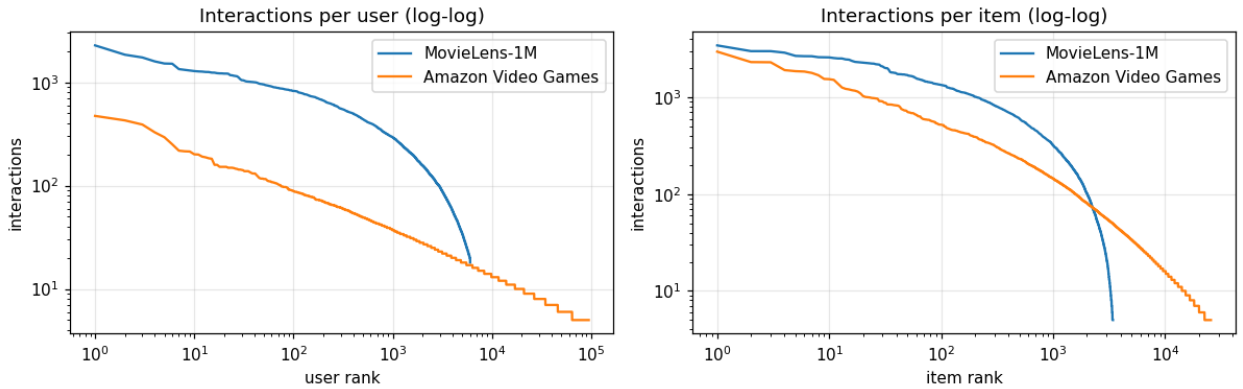


Figure 1: Interactions per user (left) and per item (right), log-log.

3.2 Protocol

We evaluate by *full ranking*: for each user, all items not seen in training are scored and Recall@K / NDCG@K ($K \in \{10, 20\}$) are computed on the held-out items; no sampled-negative evaluation is used, as it is known to bias comparisons. Model selection uses validation NDCG@10 with early stopping (patience 30 epochs). The training budget is 300 epochs for MF (which always early-stops far below it) and 600 for LightGCN; the reason for the larger budget is discussed in Section 4.3. The full queue of 18+ runs completes in ≈1.5 h on one NVIDIA L4 GPU.

The main experiments use the standard per-user *random* 80/10/10 split (as in the LightGCN paper). Because random splits let a model "see the future" — including the future popularity of items — we additionally re-run all four methods under a stricter *temporal* split (per user, the chronologically last 10% of interactions form the test set) as a robustness check in Section 4.7.

4 Results and Analysis

4.1 Main comparison

Dataset	Model	R@10	N@10	R@20	N@20	Best ep.
MovieLens-1M	Popularity	0.0731	0.1304	0.1223	0.1329	–
	ItemKNN (k=100)	0.1323	0.2199	0.1961	0.2188	–
	MF-BPR	0.1560	0.2456	0.2384	0.2525	45
	LightGCN (L=3)	0.1728	0.2605	0.2638	0.2709	450
	LightGCN vs. MF	+10.8%	+6.1%	+10.6%	+7.3%	
Video Games	Popularity	0.0267	0.0136	0.0426	0.0177	–
	ItemKNN (k=100)	0.0636	0.0380	0.0885	0.0445	–
	MF-BPR	0.0634	0.0344	0.0969	0.0431	125
	LightGCN (L=3)	0.0907	0.0494	0.1349	0.0608	565
	LightGCN vs. MF	+43.2%	+43.5%	+39.3%	+41.2%	

Table 2: Test-set results (R = Recall, N = NDCG; d=64 for the learned models). The method generations improve in order on both datasets; LightGCN wins everywhere, but its gain over MF is ∼7× larger on the sparse dataset.

Table 2 shows the central result. The four methods line up in generation order on both datasets — Popularity < ItemKNN < MF-BPR < LightGCN — and the baselines calibrate the scale: on MovieLens-1M the no-personalization Popularity list already reaches roughly half of MF's NDCG@10, so the interesting differences are concentrated in the upper half of the table. On dense MovieLens-1M, LightGCN improves NDCG@10 over MF by +6.1%; on Video Games the same architecture change yields +43.5%. The MF/LightGCN comparison isolates the propagation step as the only difference, so we conclude that **the value of higher-order graph signal grows sharply as direct per-user signal shrinks**. On dense data most of what the graph could tell the model is already present in each user's own long history.

4.2 Convergence behaviour

Figure 2 shows the cost of the graph: LightGCN needs both a full-graph propagation per optimization step *and* many more epochs. On Video Games the two curves cross around epoch 90 — before that,

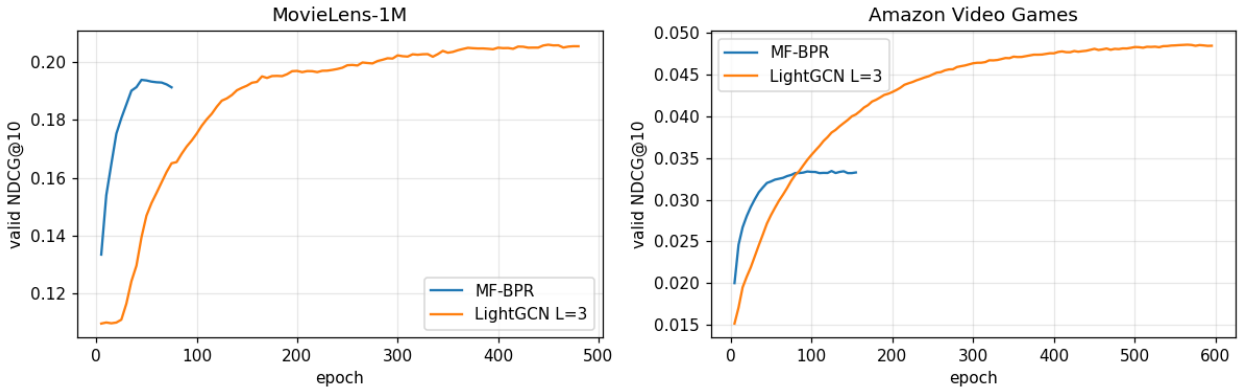


Figure 2: Validation NDCG@10 per epoch. MF converges fast and early-stops (best epoch 45/125); LightGCN converges ∼10× slower (best epoch 450/565) but reaches a higher plateau.

MF is the better model. In compute-constrained settings this trade-off matters as much as the final number.

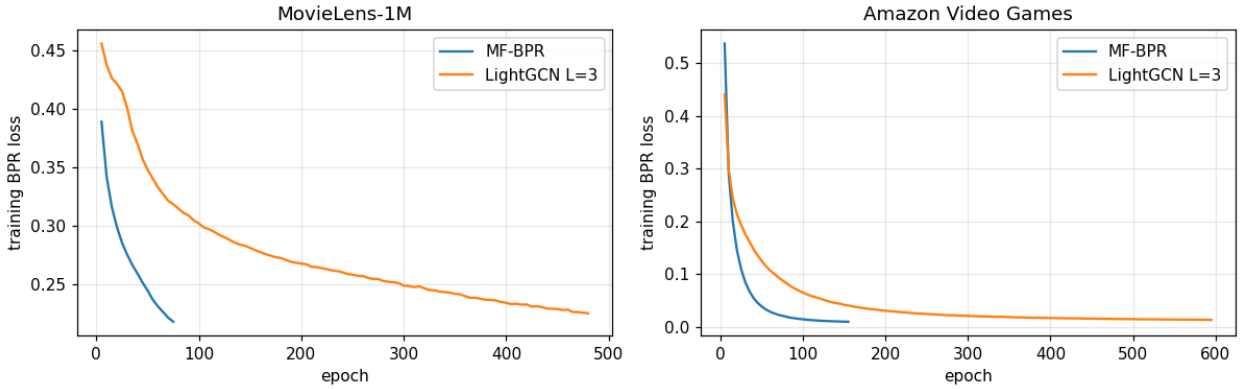


Figure 3: Training BPR loss per epoch. MF drives the loss down faster and further (its curves stop early because of early stopping), yet ranks held-out items worse.

The training-loss curves (Figure 3) complete the picture from the optimization side: MF fits the training pairs faster and more aggressively than LightGCN, yet generalizes worse (Table 2). The propagation step acts as a structural regularizer — each embedding is tied to its neighborhood average, so LightGCN cannot minimize the loss by memorizing individual pairs. The gap between training fit and test ranking is precisely the graph prior at work, and it also explains the slower convergence observed above.

4.3 Ablation: propagation depth, and a training-budget pitfall

The layer count L is LightGCN's key component: L=0 recovers MF, so Figure 4 directly measures the marginal value of each propagation hop. Two observations. First, a single hop already captures most of the gain on the sparse dataset (+26% NDCG@10 over MF), with diminishing returns after L=3. Second — and this is a finding we consider as valuable as the curve itself — our *initial* run of this ablation used a 300-epoch budget for every L and showed the classic "peak at shallow L, degradation at L=4" pattern commonly attributed to over-smoothing. Doubling the budget

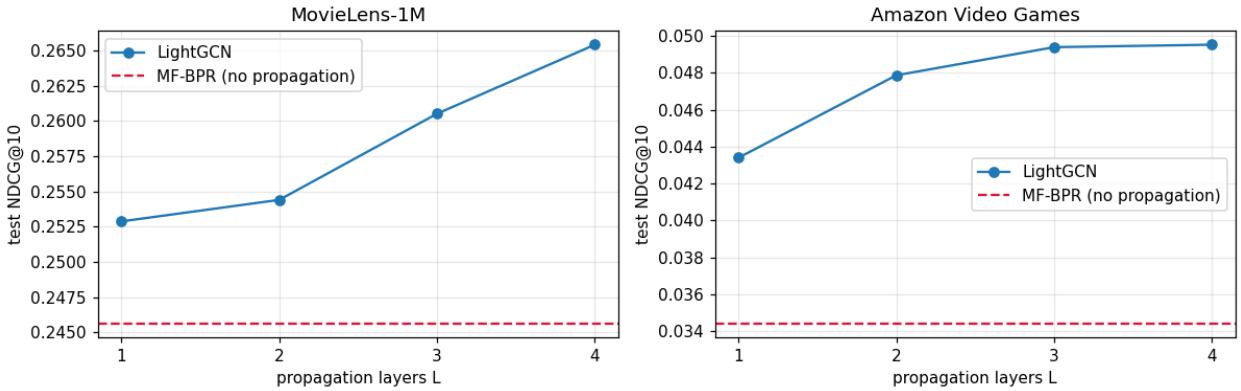


Figure 4: Test NDCG@10 vs. propagation layers L, all configurations trained with the same fair 600-epoch budget. Even L=1 clearly beats MF; no over-smoothing is visible up to L=4.

removed the degradation entirely: on MovieLens-1M the best epoch grows from 110 at L=1 to ∼565 at L=4 (170 to ∼555 on Video Games), i.e. **deeper propagation converges more slowly, and an unfair training budget masquerades as over-smoothing**. Ablation tables in the literature that fix a single epoch budget across depths should be read with this caveat in mind.

4.4 Embedding dimension

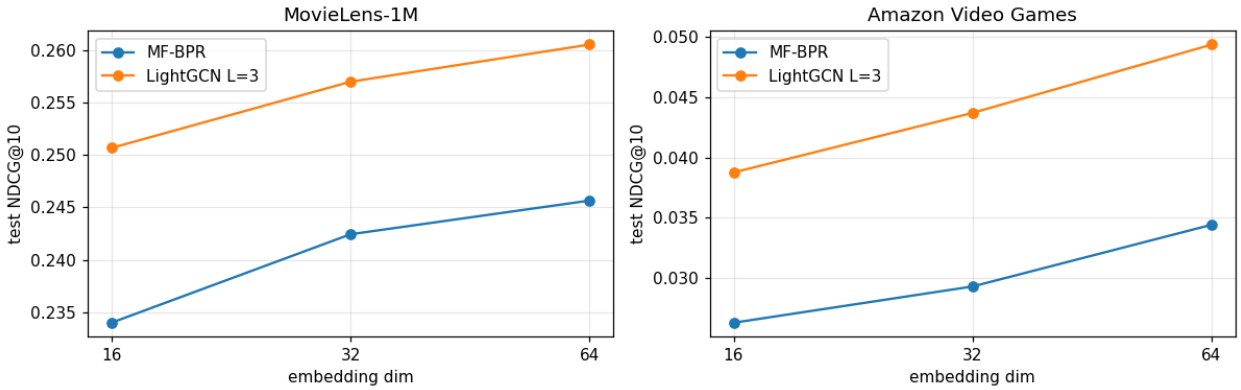


Figure 5: Test NDCG@10 vs. embedding dimension $d \in \{16, 32, 64\}$.

Both methods improve monotonically with d (Figure 5), and the LightGCN curve is shifted upward roughly in parallel. Notably, **LightGCN at d=16 beats MF at d=64 on both datasets** despite having a quarter of the parameters: the graph prior contributes more than raw capacity does in this regime.

4.5 Who benefits? Per-user analysis

Bucketing test users by their training degree (Table 3, Figure 6) explains where the improvement lives: on MovieLens-1M the gain falls monotonically from +14.6% for the coldest quartile to +4.0% for the most active one. Propagation lets a cold user borrow the histories of neighboring users; a heavy user's own history already positions them well, and the borrowed signal adds little. A case

Dataset	Train inter.	MF-BPR	LightGCN	Gain
MovieLens-1M	14–35	0.2877	0.3299	+14.6%
	36–75	0.2711	0.3055	+12.7%
	76–164	0.2257	0.2442	+8.2%
	≥165	0.1707	0.1775	+4.0%
Video Games	3–3	0.0932	0.1335	+43.3%
	4–6	0.0962	0.1348	+40.1%
	≥7	0.1020	0.1367	+34.0%

Table 3: Recall@20 by user-activity bucket (number of training interactions). The relative gain shrinks monotonically as activity grows.

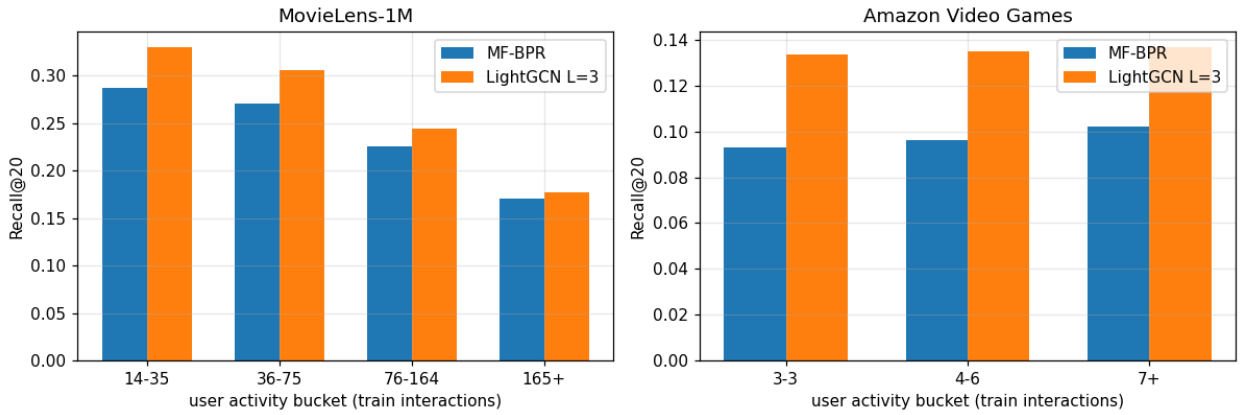


Figure 6: Recall@20 per activity bucket, corresponding to Table 3.

study makes this concrete (notebook, Section 6.5): for a sampled 476-interaction user, MF places *seven* held-out movies in its top-10 versus LightGCN's four — averaging in neighborhood signal can actually dilute an already well-determined profile.

4.6 Popularity bias and catalog coverage

Dataset	Model	Mean popularity	Coverage@10
MovieLens-1M	MF-BPR	1150	50.9%
	LightGCN	994	61.4%
Video Games	MF-BPR	373	62.1%
	LightGCN	418	41.4%

Table 4: Mean training popularity of top-10 recommendations and share of the catalog that ever appears in a top-10 list.

Table 4 reveals that *how* LightGCN wins differs by regime. On dense MovieLens-1M it recommends *less* popular items than MF (mean popularity 1150→994) and covers *more* of the catalog (51%→61%): the rich graph individualizes users. On sparse Video Games the opposite happens: propagation over a thin graph concentrates mass on bestsellers (373→418) and coverage drops from

62% to 41%. Part of the +43.5% headline gain is therefore a *popularity bet*. If the product goal values discovery and long-tail exposure, this is a genuine weakness of LightGCN in sparse regimes — and, we believe, the most promising direction for improving the method.

4.7 Robustness check: temporal split

		Random split		Temporal split	
Dataset	Model	N@10	R@20	N@10	R@20
MovieLens-1M	Popularity	0.1304	0.1223	0.0635	0.0686
	ItemKNN (k=100)	0.2199	0.1961	0.0812	0.0971
	MF-BPR	0.2456	0.2384	0.0760	0.0995
	LightGCN (L=3)	0.2605	0.2638	0.0763	0.0990
	LightGCN vs. MF	+6.1%	+10.6%	+0.4%	-0.6%
Video Games	Popularity	0.0136	0.0426	0.0129	0.0381
	ItemKNN (k=100)	0.0380	0.0885	0.0192	0.0511
	MF-BPR	0.0344	0.0969	0.0232	0.0644
	LightGCN (L=3)	0.0494	0.1349	0.0321	0.0882
	LightGCN vs. MF	+43.5%	+39.3%	+38.2%	+36.9%

Table 5: Random vs. temporal (per-user leave-last-10%) splits, all four methods. Absolute numbers drop sharply under the temporal protocol; the LightGCN-over-MF gain survives on the sparse dataset but vanishes on the dense one.

Random splits let a model "see the future", including the future popularity of items, so we re-ran all four methods under a strict chronological split (Table 5). Three observations. First, *every* method loses most of its absolute score (e.g. MF on MovieLens-1M: NDCG@10 0.246→0.076) — a measure of how much random-split numbers are inflated by temporal leakage, and a reminder that offline numbers under the standard protocol must not be read as deployment estimates. Second, and central to our thesis: **the sparse-data advantage survives the strict protocol essentially intact (+43.5% → +38.2% NDCG@10), while the dense-data advantage disappears (+6.1% → +0.4%)**. The small dense-data gain observed under the random split falls within the margin attributable to leakage; the sparse-data gain does not. Our headline conclusion therefore sharpens: graph propagation helps *where data is sparse*, and only there does the claim withstand the stricter protocol. Third, on MovieLens-1M under the temporal split the non-learned ItemKNN (0.0812) outperforms both learned models (≈0.076) — when the test set is literally "what the user watched next", recent co-occurrence patterns beat static embeddings, pointing toward temporally-aware models as the proper tool for that regime (out of scope here).

4.8 Extension: degree-α normalization against sparse-regime popularity bias

Sections 4.7 and the coverage analysis identified LightGCN's weakness in the sparse regime: propagation concentrates probability mass on popular items. We propose a one-line extension that attacks this at the mechanism level. In LightGCN a message from node s to node t is weighted $(d_s d_t)^{-1/2}$; we generalize this to

$$w_{s\to t} = d_s^{-\alpha}\, d_t^{-(1-\alpha)}, \tag{3}$$

where α=0.5 recovers the original model exactly (another clean ablation by construction) and larger α makes high-degree (popular) nodes "speak more quietly" during propagation. Related

normalization families have been explored for the accuracy–novelty trade-off [6]; our contribution is to connect the knob directly to the sparse-regime coverage collapse we measured, and to evaluate it there.

Model	N@10	R@20	Coverage@10	Mean pop.
MF-BPR	0.0344	0.0969	62.1%	373
LightGCN α=0.2	0.0497	0.1374	57.2%	378
LightGCN α=0.3	0.0508	0.1390	53.4%	406
LightGCN α=0.4	0.0504	0.1380	47.4%	426
LightGCN α=0.5 (original)	0.0494	0.1349	41.4%	418
LightGCN α=0.6	0.0468	0.1307	30.0%	469
LightGCN α=0.7	0.0441	0.1246	25.0%	446
LightGCN α=0.8	0.0420	0.1198	15.3%	486

Table 6: Degree-α sweep on Video Games (L=3, d=64, 600-epoch budget). Accuracy and beyond-accuracy metrics of the recommendation lists.

Table 6 reports accuracy together with catalog coverage and the mean training popularity of recommended items, and it contains a lesson in both directions. Our *initial* hypothesis was that $\alpha > 0.5$ (silencing popular *senders*) would debias the model; the experiment refuted it monotonically — coverage collapses from 41.4% to 15.3% as α rises to 0.8, with accuracy falling alongside. The mechanism: raising α necessarily lowers the *receiver* exponent $(1-\alpha)$, so popular items accumulate mass with less normalization when receiving messages, and this effect dominates. The true debias direction is therefore $\boldsymbol{\alpha < 0.5}$: normalize receivers harder. There, α=0.3 **is a strict Pareto improvement over the original LightGCN**: NDCG@10 rises 0.0494→0.0508 (+2.8%), Recall@20 +3.0%, catalog coverage recovers from 41.4% to 53.4% (of the 21 points lost relative to MF, more than half is regained), and the mean popularity of recommendations drops below the original's. At α=0.2 coverage climbs further to 57.2%, approaching MF's 62.1%, while accuracy begins to fall back toward the original, locating the accuracy peak at $\alpha \approx 0.3$. A single mechanism-level knob thus fixes most of the weakness identified in Section 4.6 *while improving accuracy*, and the refuted first guess is itself informative: in bipartite propagation, how strongly popular nodes *receive* matters more than how loudly they *speak*.

5 Discussion: strengths, weaknesses, key factors

MF-BPR is fast (best epoch within 45–125), simple, trivially scalable, and surprisingly strong on dense data and heavy users; its weakness is cold users, for whom it has almost no signal. **LightGCN** delivers consistent accuracy gains that concentrate exactly where MF is weakest, and is remarkably parameter-efficient; its costs are $\sim$10$\times$ slower convergence, a full-graph propagation per step, and a popularity/coverage regression on sparse graphs. The key factors affecting performance in our experiments were, in order: dataset density, training budget (especially for deeper propagation), propagation depth L, and embedding dimension. Sensible defaults from our sweeps: L=3, d=64, with the caveat that L and the epoch budget must be increased together.

6 Conclusion

We compared MF-BPR and LightGCN under a controlled protocol in which graph propagation is the only moving part, anchored by popularity and ItemKNN baselines. The higher-order signal matters where direct signal is scarce: sparse datasets and cold users. Our temporal-split check makes this precise — the sparse-data gain survives the strict protocol (+38.2%), while the dense-data gain proves to be largely an artifact of random-split leakage (+0.4%). The gains are also not free: slower convergence, more compute per step, and on sparse data a shift toward popular items that sacrifices 21 points of catalog coverage. Two methodological lessons generalize beyond this project: training budget can masquerade as over-smoothing in depth ablations, and split protocol can masquerade as model superiority — ablations must equalize convergence, and headline claims should be re-checked under temporal splits. Finally, acting on our own diagnosis, we implemented a popularity-debiasing extension — the degree-α normalization of Section 4.8 — and found that α=0.3 improves accuracy and catalog coverage simultaneously in the sparse regime, confirming that the popularity concentration we measured was not a necessary price of graph propagation but an artifact of its symmetric normalization.

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